MANDOWN V6

Asymmetric-Posture Geometric Fix, Temporal Motion-Signature Confirmation & Detection-Recovery — Implementation, Real-Image Validation & Honest Findings
Prepared: September 2026 | Validated against: 7 real live-camera screenshots (no live feed available this round) + the 4 standing reference videos

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1. Executive Summary

This round targeted two problems left open by the V5 report: (1A) a foreshortened-lying person whose limbs are posed *asymmetrically* (a raised knee, a raised arm, an arm tucked under the torso) still doesn't reliably confirm, because every existing angle feature averages left/right keypoints into one midpoint — an implicit symmetry assumption that breaks exactly for this posture; and (1B) the pose model itself sometimes produces no detection at all for a person already on the floor, undercounting a visibly-present fallen person.

Three fix sets were implemented, all gated to be either provably monotonic with respect to every existing confirmed case, or purely additive/fallback-only, so the single most protected real regression case in this project (`M_D_2.mp4`) could not be made worse by construction. This report documents exactly what was built, what real measurement showed, and — matching this project's established discipline of recording negative results, not just successes — **an important honest finding: the specific visibility-asymmetry mechanism Fix Set A targets does not actually explain the real §7.1 gap once re-measured against the real MediaPipe output for that exact frame**. The fix is safe and kept (it cannot regress anything and may help other real cases), but it did not close the original target case on its own. Fix Set B (a temporal, not geometric, confirmation path) is the more promising path for that specific gap, but by its own nature can only be validated on real video with genuine motion — untestable on the static images available this round.

A serious-looking regression was also investigated this round: a fresh run of `M_D_2.mp4` today produced **0 confirmed falls** instead of the previously-reported 1. After a careful investigation (§7), this was traced to this video's own already-documented tracker-fragmentation behavior (75-80 track IDs for one physically continuous person), not to anything in this round's code changes — but it means the literal "exact-match" re-verification requested could not be produced this round, and this is reported transparently rather than papered over.

2. Problem Restated: Two Independent Failure Modes

2.1 (1A) Asymmetric foreshortened-lying postures

Every angle feature in the pre-V6 pipeline (`torso_angle` , `hip_angle` , the 3D-lifted `torso_angle_3d`) is computed from the *midpoint* of the left/right shoulder or hip pair, or the midpoint-derived 3D center MediaPipe infers. Averaging two limbs into one number is an implicit symmetry assumption. Foreshortened lying is exactly the case where that assumption breaks: one limb reads unambiguously "lying," the other reads "bent/raised," and the average sits in the ambiguous middle where nothing confirms — documented in the V5 report as the open §7.1 case (a person lying on their back with one knee raised): `torso_angle_3d=0.319` (below the 0.55 floor), `down_score=0.454` (below the 0.65 floor), `floor_proximity=0.552` (below the 0.70 floor).

2.2 (1B) The pose model itself sometimes fails to produce a detection at all

YOLO-pose, like nearly every pose model, is trained overwhelmingly on upright/walking/sports imagery. Prone, floor-level, foreshortened bodies are underrepresented in that distribution, so both detection confidence and keypoint quality degrade on exactly the frames MANDOWN most needs to see clearly — a data-distribution problem in the frozen model, worked around at inference time (not fixed by retuning fusion thresholds, and not by retraining).

3. Fix Set A — Per-Side (Asymmetric-Aware) Geometric Features

3.1 §2.1 — Per-leg `leg_extension_ratio`

Investigation found this selection rule was **already implemented**, independently, in an earlier engineering round: `ExtendedFeatureExtractor._leg_extension_ratio` computes each leg's hip→knee→ankle extension ratio separately and only includes a leg whose three keypoints are all confidently visible; when exactly one leg qualifies, that leg's ratio is used alone (never blended with a low-confidence value); when neither qualifies, it returns a neutral "no bend evidence" default rather than fabricating one. This is functionally identical to the V6 spec's requested selection rule. New tests (`TestLegExtensionRatioPerSideSelection`, in `tests/test_extended_features.py`) now assert this exact behavior explicitly, closing the gap between "this happens to already work" and "this is verified to work."

No threshold value was changed here: the spec's suggested τ_vis=0.5 (vs. the existing 0.3) was deliberately **not** adopted, since the spec's own guidance explicitly warns against tuning this in isolation from a large real keypoint-confidence dataset, which this round did not have (no live feed).

3.2 §2.2 — Per-side `torso_angle_3d`, provably monotonic via `max()`

Implemented in `src/pose_3d.py`. When one side's shoulder+hip MediaPipe landmarks are confidently more visible than the other's by more than `Pose3DConfig.torso3d_asymmetry_threshold` (0.3, a design margin — the spec's own suggested starting point), that side's *own* shoulder→hip vector is computed as a second candidate angle, and the **larger** of it vs. the existing bilateral (midpoint) reading is kept:

```
torso_angle_3d = max(bilateral_angle, single_side_angle)   # only when |w_L - w_R| > threshold
                                                             # else:
bilateral_angle, byte-identical to pre-V6
```

> **Why this is provably monotonic:** a symmetric-visibility body (the common case, including every existing confirmed real fall in this repo) takes the unchanged bilateral-only path — zero behavior change. An asymmetric-visibility body can only have its angle *raised* relative to the pre-V6 value, never lowered, since `max()` never selects a smaller number over the one that was already there. This means the fix cannot regress any existing confirmed case by mathematical construction, independent of any live-camera re-verification.

Verified with 5 new unit tests (`TestPose3DPerSideMonotonicMax`, `tests/test_pose_3d.py`) using a mocked MediaPipe result for precise control: symmetric visibility stays byte-identical to bilateral; an asymmetric fixture modeled on the real §7.1 corruption mechanism (a raised knee pulling one hip

toward the shoulder) shows the single-side reading correctly wins via `max()`; a visibility gap below the threshold stays bilateral (no over-triggering on ordinary noise); a genuinely upright person with incidental visibility asymmetry stays correctly low (both readings are low, so `max()` changes nothing).

3.2.1 Real-image re-measurement — the honest finding

Re-running this exact fix against the real §7.1 frame (2026-09-14 18:19:32 screenshot) to see whether it actually raises `torso_angle_3d` for the real target case produced a genuinely important, unglamorous result:

```
L_SHOULDER visibility=0.999   L_HIP visibility=0.998
R_SHOULDER visibility=0.999   R_HIP visibility=0.997
w_L = 0.997   w_R = 0.997   |w_L - w_R| = 0.0002   (threshold: 0.3 -- NOT crossed)
=> per-side path never activates; result stays bilateral-only: torso_angle_3d =
0.236
```

> **MediaPipe is confidently wrong here, not uncertain.** The hypothesis behind Fix 2.2 was that a raised knee drags visibility DOWN on the corrupted side, creating a detectable asymmetry. On this real frame (and on 2 other real lying postures re-measured the same way — the "arm raised" and "symmetric lying" screenshots, see §6), MediaPipe reports **near-identical, high visibility on both sides** even though its inferred 3D position is apparently still off for this posture. The corruption here is a genuine 3D-pose-inference inaccuracy for an underrepresented posture, not a visibility-detectable asymmetry — so the specific mechanism this fix targets does not explain, and does not close, the original real gap it was written for.

The fix is kept anyway: it is safe (provably monotonic, independently unit-tested), free (near-zero extra cost — the extra angle computation is a few vector operations, not a second model inference), and may still help a different real scenario where visibility genuinely does diverge (e.g., a limb entirely outside the frame or heavily occluded by furniture, rather than merely differently posed). It is reported as "implemented and safe, did not close the target case" rather than overclaimed — the same honesty standard the V4.7/V4.9 investigations in the V5 report were held to.

3.3 §2.1's `hip_angle` per-side replacement — deliberately NOT implemented this round

The spec also requested a per-side, asymmetric-aware `hip_angle`. This was evaluated and **not implemented**, for a specific, provable safety reason rather than a time constraint: `hip_angle` currently feeds a *subtractive* exclusion gate — the V4.9 `hip_torso_mismatch` signal (`mismatch = torso_angle - hip_angle`) that protects against the real deep-bend false positive documented in the V5 report. Raising `hip_angle` for an asymmetric case would *shrink* that mismatch, making the bend-exclusion signature *less* likely to fire — i.e., a change that helps one real case could simultaneously weaken the protection for a different, already-fixed real false positive. Unlike `torso_angle_3d` (where "higher is always more like lying, so raising it is always the safe direction"), `hip_angle` sits on both sides of this tension at once, so no single `max()` / `min()` trick makes it safe in both directions

simultaneously without a live-footage re-verification pass this round did not have access to. Deferred, and flagged as the top open item in §9 — this mirrors the V5 report's own precedent of reverting/deferring a change the moment its safety could not be proven, rather than shipping something merely "probably fine."

4. Fix Set B — Motion-Signature Confirmation Path

A purely **temporal**, not geometric, confirmation path: `collapse_then_stillness`, computed per track in `MandownTrackContext` and consumed as a new, additive rule 8 in `EvidenceFusion`. A hard 0/1 gate (never blended), True only when **all** of:

1. a genuine `body_height_norm` collapse of at least `theta_spike` within a 1-second window, somewhere in the last `t_lookback_seconds + t_still_seconds`;
2. continuous stillness (per-frame keypoint motion energy below `theta_still`) held from that spike through to now, for at least `t_still_seconds`;
3. a real, currently-elevated `height_collapse_cue` (reuses the same calibrated floor established in V4.2-V4.9);
4. the track is not currently inside its own confidently-standing baseline.

```
if collapse_then_stillness and pose_quality >= pose_quality_floor:
    fusion_score = max(fusion_score, confirmed_mandown_threshold)   # same style as
V4.10's direct down_score path
```

This is additive by construction — it can only ever raise the fusion score for a track already backed by a real, sustained height collapse; it cannot fire for, and cannot suppress, anything else.

4.1 Thresholds measured from real footage, not guessed

Per the spec's own instruction, `theta_spike` / `theta_still` were measured from this repo's 2 real single-fall videos via a dedicated diagnostic pass (not a live camera — these are pre-existing repo assets):

Video / track	1s collapse velocity at real fall onset	Post-onset "settled" motion-energy (p95)
`mandown_2.mp4` track 1	**0.560**	**0.012**
`man_down_c08.mp4` track 1	**0.137**	**0.053**

`theta_spike=0.12` sits just below the weaker of the two real onsets; `theta_still=0.06` sits just above the higher of the two real "settled" readings (real footage is never perfectly motionless — ongoing minor jitter/breathing/pose-estimation noise is normal, even while genuinely lying still). Explicitly labeled in-code and here as a **conservative first pass from a small (n=2) real sample**, not a fully-tuned production threshold — set with margin toward not falsely triggering, since every real confirmed case in this repo already confirms through an existing path independent of this rule ever firing.

4.2 Unit-tested negative controls (per spec §3.4)

Negative control	Which condition fails	Result
Deliberate, gradual sit-down (height declines slowly over 10s, not sharply within 1s)	(a) no genuine spike	Correctly rejected
Chair scoot-back (height never really collapses)	(c) height_collapse_cue stays low	Correctly rejected
Bending to help someone, then standing back up (motion resumes 2s after the spike, well before the 5s stillness window completes)	(b) stillness doesn't sustain	Correctly rejected
Track currently reading confidently upright	(d)	Correctly rejected
Freshly-created track, no history	insufficient data	Correctly rejected, no crash

4.3 What could and could not be validated without live video

Per the spec's own explicit acknowledgment, the TRUE-POSITIVE path is inherently temporal and cannot be exercised on a static image. What *was* run: feeding one of the real still images repeatedly as a synthetic zero-motion, zero-height-change 200-frame "sequence" —

```
PASS: collapse_then_stillness stayed False for 200 static-repeat frames (no genuine spike ever occurred)
```

This proves the gate does not false-trigger on a static/frozen feed, but is **not** a substitute for validating the true-positive path on a real fall clip — that remains deferred to when live-feed testing resumes, exactly as the spec anticipated.

5. Fix Set C — Detection-Recovery for Missed Poses

5.1 §4.1 — Narrow-range rotation search

Implemented in `src/detector.py` as `PersonDetector.redetect_in_roi_with_rotation`, run **only** when the existing plain straight-crop re-acquisition (`redetect_in_roi`, unchanged) has already failed to find a previously-tracked person at all — i.e., strictly rarer than the spec's own proposal (which would try rotation alongside the plain retry on every trigger). Tries {prev_best-15°, prev_best, prev_best+15°} on the same padded ROI using the *same frozen model* (no retraining), keeps whichever candidate gives the highest mean keypoint confidence, and decays the remembered best angle back toward 0 (×0.9/frame) when not re-triggered.

Mechanism-level validation (real image)

Both real §7.1-adjacent fixtures were already confidently, correctly detected in their original frames (det. confidence 0.827 and 0.909) — i.e., Problem 1B does not manifest in these exact two frames, so no "before/after recovery" could be demonstrated on them specifically (an honest limitation, reported rather than manufactured). A direct mechanism check was run instead: the real standing person from the arm-raised screenshot was synthetically rotated by known amounts and re-detected:

Synthetic tilt	Plain straight-crop mean keypoint conf.	Rotation-search mean keypoint conf.
15° (within the ±15° search window)	**0.774**	**0.811 (won at +15°)**
45° (exceeds the narrow search window)	**0.826***	**0.826 (won at 0° — correctly could not reach 45°)**

*YOLO-pose was itself fairly robust to the 45° case here on this particular crop (padding_ratio=0.6 retains enough context) — a real, modest improvement was measured within the search window, and the boundary case correctly shows the narrow-search design limiting itself rather than silently failing.

5.2 §4.2 — Confidence-decayed keypoint carry-forward

Implemented as `MandownTrackContext.extrapolated_keypoints`: constant-velocity extrapolation from the track's last 2 confident raw keypoint arrays, with confidence decaying by `confidence_decay_factor` (0.85) per frame since the last real detection, capped at `t_gap_max_seconds` (3.0s) of continuous non-detection before the track is dropped for real. Wired as a last-resort fallback in `_reacquire_lost_tracks` — only when *both* the plain retry and the rotation search have failed — and fed into `frame_result.persons` exactly like a real detection, so the existing `pose_quality_score` discount handles decaying trust with zero new fusion-side code, per the spec's own composability goal.

> As the spec itself anticipates, this path is **untestable on a single static image** (it requires an actual multi-frame detection gap). It is unit-tested at the mechanism level (constant-velocity extrapolation math, confidence decay, insufficient-history graceful failure — `TestExtrapolatedKeypoints`), but its real-world behavior on an actual brief occlusion is deferred to live-feed testing, exactly as instructed.

6. Real-Image Validation Results (7 Screenshots)

The 7 screenshots referenced in the task (found on disk, matching every timestamp given) were run through the real single-frame pipeline (YOLO-pose → `ExtendedFeatureExtractor` → `Pose3DEstimator` → `FeatureMandownScorer` → `EvidenceFusion`), with **no temporal history and no LSTM** (a single isolated frame has neither) and, since a desktop screenshot's window-chrome offset cannot be reliably mapped back onto this camera's saved 1280×720 perspective calibration, a **same-image height reference** (the median bounding-box height of the other confidently-standing people visible in the same screenshot) substituting for the real per-camera 2D calibration used in production. This is explicitly a cruder, single-frame approximation — labeled as such here and in the diagnostic script — not the real calibrated pipeline.

6.1 Positive fixtures

Frame	Person	down_score	floor_prox	torso_angle_3d	height_collapse_cue	fusion / tier
2026-09-14 18:19:32 (prone, arms under torso, one knee bent)	fallen person	**0.068**	**0.164**	**0.236**	**0.148**	**0.026 / NORMAL**
2026-09-03 17:49:50 (arm raised toward ceiling)	fallen person	**0.121**	**0.341**	**0.494**	**0.356**	**0.051 / NORMAL**

Neither reached a confirm-eligible instantaneous score. This is consistent with §3.2.1's finding (the per-side 3D fix doesn't engage here) and with the fact that `persistence_score=0.0` on a single isolated frame — the geometric-alone confirm path requires sustained persistence (0.65+), which no single frame can ever provide by definition. The one path that does *not* require persistence — a very high `down_score` alone (≥0.65) — was not reached on either real frame under this crude same-image calibration. This is an honest, negative instantaneous-validation result, reported plainly rather than reframed as a success.

6.2 Negative-control fixtures (must NOT confirm)

Frame	# standing/walking people checked	Highest fusion score among them	Result
2026-09-03 17:58:39 (standing + walking)	1	**0.028**	Correctly NORMAL

2026-09-03 17:58:46 (standing)	2	**0.152**	Correctly NORMAL
2026-09-03 17:49:50 (bystander standing/bending over phone)	2	**0.184**	Correctly NORMAL

No new false positive was introduced among any standing/walking/bending person visible across all 7 images — consistent with the fact that every V6 change is either provably monotonic (Fix A/B) or fallback-only and re-acquisition -scoped (Fix C), none of which can manufacture evidence for a person who genuinely isn't down.

6.3 The two "not yet detected" symmetric-lying frames

Two further real frames (2026-09-03 17:31:36 and 17:32:51) show the same person lying symmetrically (arms folded at chest) on the floor near the bar counter, both reading `Falls: 0` in the live app at the time. Re-measured this round: `torso_angle_3d=0.596-0.638` (correctly, confidently reads as lying — MediaPipe itself is accurate for this particular symmetric posture), but `down_score` / `height_collapse_cue` stayed low without real calibration. A third frame minutes later (17:39:16), same person now in a more diagonal pose (one arm and leg extended away from the body), **was** already correctly detected by the live app at the time (`Falls: 2`) — consistent with the pre-V6 2D alternate-grounding path already handling genuinely diagonal lying, while the more foreshortened/symmetric earlier poses were the harder case this round targeted.

7. Regression Verification: The 4 Reference Videos

Full test suite: **266 passed, 0 failed** (up from V5's 250 — 16 new tests added this round, zero regressions in the existing suite). The real-video re-verification below was run today via the actual `FallDetectionPipeline` (not a diagnostic approximation), per the spec's explicit shippability gate.

7.1 Clean (lightly-fragmented) videos — exact or near-exact match

Video	V5-reported confidence	V6 re-run confidence	Persons tracked	Result
`mandown_2.mp4`	**0.881**	**0.881**	3 (was 3)	Exact match
`man_down_c08.mp4`	**0.841**	**0.836**	1 (was 1)	Near-exact (Δ0.005, within run-to-run GPU/tracker noise)
`man_down_zoomed_demo__2_.mp4`	**0.867**	**confirmed, 1/1**	8 (was 9)	Confirmed, zero new false positives among the rest

7.2 `M_D_2.mp4` — the protected track: NOT reproduced this round

> **Two fresh, full runs of `M_D_2.mp4` today (with, and separately without, Fix Set C entirely disabled) both produced 0 confirmed MANDOWN events**, versus the 1 confirmed event (confidence 0.831, down_score 0.813) documented in the V5 report. This is reported transparently, not minimized.

Investigation

- **Track fragmentation, not scoring failure:** both runs show 75-80 total tracked persons for this one video (vs. 1-8 for the other 3 reference videos) — consistent with this project's own pre-existing, already-documented observation (`TrackBridgeConfig` 's own code comment: "one real validation video went from a handful of tracks to 74 under current settings" — almost certainly this same video). At least 3 separate track fragments in today's run individually reached genuinely strong MANDOWN-tier evidence at their peak frame (down_score 0.643-0.808, fusion_score 0.79-0.828, `suggested_tier: "MANDOWN"`), but none individually sustained the 8 consecutive confirming frames `StateMachineConfig` requires before the fragmentation split them apart again.
- **Fix Set C ruled out:** disabling rotation search and confidence-decayed carry-forward entirely did not restore confirmation — if anything, the disabled run showed *more* fragmentation (80 tracks) than the enabled one (75 tracks), the opposite of what a Fix-Set-C-caused regression would predict.
- **Fix Set A/B ruled out mathematically:** §3.2's per-side torso_angle_3d uses `max()` and §4's collapse_then_stillness is a pure OR-additive rule — both are provably incapable of *lowering* any score for any input, so neither can be the mechanism behind a track that used to confirm no longer doing so.
- **Most likely explanation:** inherent run-to-run non-determinism in the underlying YOLO/BoT-SORT stack (unpinned CUDA op ordering is a well-known source of small run-to-run differences in deep learning inference), compounding over a long, heavily-fragmented video (3718 frames) into a different track-ID split than whichever run produced the V5 report's figures.

Conclusion, stated plainly: the literal "exact-match re-verification with zero regression" this report was asked to produce could not be achieved this round for this specific video, and no amount of re-running alone would prove it either way, since the system's fragmentation behavior for this video is not observed to be stable run-to-run even before considering any V6 change. The mathematical monotonicity of every V6 scoring change, combined with the clean exact/near-exact matches on the other 3 reference videos (which do not suffer this fragmentation), is the strongest evidence available this round that V6 itself did not cause this — but it is evidence, not proof, and is reported as an open item rather than a closed one. See §9 for the recommended next step.

8. Honest Accounting: What Did and Did Not Close the Target Gap

Item	Status	Note
Fix Set A §2.1 (per-leg `leg_extension_ratio`)	Confirmed already correct	Pre-existing; now explicitly tested
Fix Set A §2.2 (per-side `torso_angle_3d`)	Implemented, safe, did not close §7.1	MediaPipe visibility stays symmetric even when its 3D inference is wrong for this posture
Fix Set A §2.1 (per-side `hip_angle`)	Deliberately not implemented	Not provably safe against the V4.9 bend-exclusion gate without live re-verification
Fix Set B (`collapse_then_stillness`)	Implemented, thresholds measured, negative controls pass	True-positive path untestable without live video (by the spec's own admission)
Fix Set C §4.1 (rotation search)	Implemented, mechanism validated (0.774 → 0.811)	Neither real target frame exhibited Problem 1B to demonstrate an end-to-end recovery
Fix Set C §4.2 (confidence-decayed carry-forward)	Implemented, unit-tested at the mechanism level	Untestable on a static image by construction
`M_D_2.mp4` exact-match regression gate	Not reproduced this round	Traced to pre-existing tracker fragmentation, not mathematically attributable to any V6 change — see §7.2
§7.1's original real gap (bent-knee lying)	Still open	Remains the top item for live-feed re-testing — see §9

9. Files Changed / Tests Added

File	Change
`src/pose_3d.py`	Per-side, monotonic-max `torso_angle_3d` (§3.2)
`src/mandown_track_context.py`	`collapse_then_stillness`, `extrapolated_keypoints`, height/motion-energy history tracking, rotation-search angle memory
`src/fusion.py`	New additive rule 8 (`collapse_then_stillness`)
`src/detector.py`	`redetect_in_roi_with_rotation`, image rotate/unrotate helpers
`src/pipeline.py`	Wires collapse_then_stillness into fusion; extends `_reacquire_lost_tracks` with the rotation-search and extrapolation fallbacks; threads raw keypoints into `ctx.update()`
`src/config.py`, `configs/mandown_config.yaml`	New `Pose3DConfig.torso3d_asymmetry_threshold`, `MotionSignatureConfig`, `RotationSearchConfig`, `DetectionRecoveryConfig`, each threshold commented with its real/measured/design-margin provenance
`tests/test_pose_3d.py`	+5 tests: per-side monotonic-max behavior
`tests/test_fusion.py`	+3 tests: rule 8 wiring
`tests/test_mandown_track_context.py`	+8 tests: collapse_then_stillness (incl. all 3 spec negative controls) + extrapolation mechanism

Full suite: 266 passed, 0 failed (was 250 at the end of V5).

10. Recommendations

1. **Highest priority, blocking full confidence in this round:** re-run `M_D_2.mp4` a few more times (with no code changes at all) to determine whether 0-confirmed is itself the more common outcome for this video under current tracker settings, independent of V6 — this would mean the V5 report's own 1-confirmed result was the fortunate run, not the reliable one, and the real fix needed is tracker-fragmentation robustness (loosening `TrackBridgeConfig`'s IoU/recency further, or investigating why BoT-SORT loses this specific track so often), not anything in this report.
2. **Live-feed re-test of Fix Set B** against a real, deliberate fall in the bent-knee-lying posture — the only way to validate the true-positive path of `collapse_then_stillness`, and the most promising remaining avenue for closing §7.1's original gap given §3.2.1's finding that the visibility-asymmetry mechanism doesn't explain it.
3. **Live-feed re-test of Fix Set C §4.2** (confidence-decayed carry-forward) against an actual brief occlusion — untestable on any static image by construction.
4. Revisit per-side `hip_angle` (§3.3) once live footage is available to directly re-verify it doesn't reopen the V4.9 deep-bend false positive, the same discipline already applied to `torso_angle_3d`.
5. Continue this project's established validation discipline for any future change: real-measured thresholds over guessed ones, explicit negative-control tests, and a full `M_D_2.mp4` re-run before considering any scoring change complete — understanding now that this specific video's tracker fragmentation may need several repeated runs, not one, to draw a reliable conclusion.

MANDOWN V6 Implementation Report — generated from real screenshot validation, unit tests, and full reference-video re-runs. Every numeric value quoted was measured directly against real footage, real screenshots, or this project's own automated test suite at the time of writing. Where a result could not be obtained or was inconclusive (the `M_D_2.mp4` exact-match gate, the true-positive paths of Fix Sets B and C §4.2), this is stated plainly rather than inferred or assumed.